

AdVantage Marketing Solution Guide

AdVantage Marketing Campaign Performance Dashboard — Full Solution & Explanation Guide

Every step, every formula, broken down piece by piece — including WHY each decision was made

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Project Overview

Dataset: AdVantage_Marketing_RAW.xlsx — 480 raw rows of campaign performance data, intentionally messy. **Final verified**

numbers: - Total Impressions: ~11.6M - Total Clicks: ~460K - Total Conversions: ~24K - Total Spend: ~\$495,280 - Total Revenue: ~\$2.2M - CTR: ~4% - Conversion Rate: ~5% - CPC: ~\$1.08 - CPA: ~\$21.05 - ROAS: 4.45x

Purpose of This Project — What We Were Trying to Find

The scenario, in plain terms: AdVantage Marketing spends money on ads across 6 different platforms (Facebook, Google, Instagram, Email, LinkedIn, TikTok). Nobody had yet connected “how much we spend” with “how much money we actually get back” — the question was simple: **is the ad money being spent well, and if not, where should it move?**

The simple questions we wanted answered: 1. Out of everyone who sees an ad, how many actually click, and how many of those turn into paying customers? 2. Which platform gives back the most money for every dollar spent? 3. Is there a platform wasting money? 4. Where in the process do we lose the most people?

What we actually found: - Out of **11.6 million ad views**, only **460,000 people clicked (about 4%)**, and only **24,000 of those actually bought something (about 5%)**. So roughly **1 in 485 people who saw an ad ended up buying**. - Overall, for every **\$1 spent on ads, the company gets back \$4.45** — a solid return. - **Google Ads gives back the most money per dollar spent**, while **LinkedIn Ads gives back the least**, even though both get similar amounts of budget. That means moving some budget from LinkedIn to Google Ads could make the money go further. - It costs about **\$1.08 per click**, but jumps to **\$21.05 per actual sale** — showing how much harder it is to get someone to buy versus just click.

Why this project is worth showing off: it turns raw ad numbers into a clear, useful answer — not just “here are the numbers,” but “here’s exactly where to move the budget and why.” That’s the difference between reporting data and giving someone something they can actually act on.

PHASE 0: The Audit — What Was Looked For and Why

- Where the real table starts
 - Channel, Device, Region — same real value spelled differently
 - Spend stored as text mixed with clean numbers (\$, USD)
 - Conversions column mixing “N/A” text with numbers
 - **Clicks > Impressions** — a genuinely impossible tracking glitch requiring a judgment call
 - Duplicate and blank rows
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PHASE 2: Cleaning — Full List with Reasoning

Column	Fix Applied	Why
Header rows	Remove Top Rows (2), Use First Row as Headers	Title/spacer row pushed real headers down
Channel	Replace Values per spelling variant → 6 clean values	“facebook ads”/“FB Ads”/“Facebook Ads” would fragment into false-separate channels in any comparison chart
Device	Capitalize Each Word → 3 clean values	Standard casing fix

Region	Replace Values (state acronym/casing variants) → 5 clean values	Same fragmentation risk as other projects
Spend	Change Type→Text (undo auto-error) → Replace Values (\$ USD → nothing) → Change Type→Decimal Number	Classic “fix value before type” pattern
Conversions	Replace Values (“N/A” → “0”) → Change Type→Whole Number	Same pattern as NorthStar’s “two” text-in-number-column issue
Date	Cleaned to Date type	Standard pattern
Clicks > Impressions (judgment call)	Added Data Quality Flag custom column instead of deleting rows	Preserves valid Spend/Revenue data on those rows while marking the Click/Impression relationship as untrustworthy for CTR-specific analysis Deliberately sequenced BEFORE adding the Data Quality Flag column —
Blank rows	Filtered via CampaignID → Remove Empty	adding a custom column first would have put text into otherwise-blank rows, making them invisible to Remove Blank Rows (a bug hit and fixed in the NorthStar project)
Duplicates	Select all columns → Remove Duplicates	Ensures only true full-row duplicates removed

Result: 460 clean rows, verified against DISTINCTCOUNT and filter dropdown checks.

Why cleaning order mattered here specifically

This project deliberately tested a sequencing lesson learned earlier: blank rows and duplicates were removed **before** adding the Data Quality Flag column, not after. If the flag column had been added first, every row (including genuinely blank ones) would have received a “Valid” or “Invalid” text value in that column — making Remove Blank Rows unable to detect them as blank anymore, since they’d no longer be fully empty. Sequencing cleanup steps correctly, rather than just doing them in whatever order feels natural, prevents this class of bug entirely.

PHASE 3: Data Modeling

Tables built:

1. **Campaign Export** (fact table — one row per campaign-day record)

- 2. **Channels** (dimension table)
- 3. **Calendar** (dimension table)

**Quick Reference: Why These 2 Extra Tables
(Channels + Calendar)**

Table	Extracted because...	Row count check
Channels	The value repeats across MANY campaign rows (many campaigns run on “Facebook Ads”, “Google Ads”, etc.) — repetition is what makes a value worth extracting into its own dimension	6 rows, verified against <code>DISTINCTCOUNT('Campaign Export'[Channel])</code>
Calendar	Generated fresh (not extracted) to guarantee a complete date range across all of 2024, which SUMMARIZE on Date alone would not provide for any day with zero campaign activity	Full Jan-Dec 2024 range, one row per day

Same one-line rule as every prior project: extract a table when a value repeats across rows, or when date completeness is needed for time-based analysis. Campaign Export itself needed neither treatment — each row already represents one unique campaign-day record with no repeating entity to pull out.

Relationships

- Calendar[Date] → Campaign Export[Date] — 1 to many
- Channels[Channel] → Campaign Export[Channel] — 1 to many

Reminder applied proactively this time: Sort by Column (Month Name → Month Number) was set immediately after building Calendar, rather than being discovered as a bug later during report-building — a direct improvement in process speed compared to the HR project.

**PHASE 4: DAX Measures — The Funnel,
Broken Down**

Why these 10 measures specifically

This project’s measures build in a deliberate SEQUENCE, mirroring the actual marketing funnel — each stage narrows from the one before it:

Impressions → Clicks → Conversions → Revenue, with **Spend** cutting across every stage to produce efficiency ratios.

Measure	Business Question
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The 5 raw funnel

Total Impressions/Clicks/Conversions/Spend/Revenue	totals — the foundation everything else is built from
CTR	“Of everyone who SAW the ad, what % clicked?”
Conversion Rate	“Of everyone who CLICKED, what % actually converted?”
CPC	“How much do we pay per click?”
CPA	“How much do we pay per actual conversion?”
ROAS	“For every \$1 spent, how many \$ did we get back?” — the headline metric

Measures 1-5: The Raw Totals

```
Total Impressions = SUM('Campaign Export'[Impressions])
Total Clicks = SUM('Campaign Export'[Clicks])
Total Conversions = SUM('Campaign Export'[Conversions])
Total Spend = SUM('Campaign Export'[Spend])
Total Revenue = SUM('Campaign Export'[Revenue])
```

All five use plain SUM() — no CALCULATE or row-by-row math needed, since each column already holds one clean number per row with nothing conditional to filter or calculate first. This is the same “when the data’s already in the right shape, use the simplest tool” principle from the HR project’s Average Salary measure.

Measure 6: Click-Through Rate (CTR)

```
Click-Through Rate (CTR) =
DIVIDE([Total Clicks], [Total Impressions], 0)
```

Piece by piece: - Numerator: how many people clicked -
Denominator: how many people saw the ad in the first place - The result answers “what fraction of viewers became clickers” — the FIRST narrowing point in the funnel

Measure 7: Conversion Rate

```
Conversion Rate =
DIVIDE([Total Conversions], [Total Clicks], 0)
```

Same DIVIDE pattern, one stage further down the funnel — of the people who already clicked, what fraction went on to actually convert (purchase/signup/etc).

Why CTR and Conversion Rate use DIFFERENT denominators: this is the core logic of funnel math — each stage’s rate is calculated against the PREVIOUS stage’s total, not against the very top of the funnel every time. CTR is “clicks ÷ impressions,” Conversion Rate is “conversions ÷ clicks” (not “conversions ÷ impressions”) — because the question being asked changes at each stage.

Measure 8 & 9: CPC and CPA

Cost Per Click (CPC) = $\text{DIVIDE}([\text{Total Spend}], [\text{Total Clicks}], 0)$
Cost Per Conversion (CPA) = $\text{DIVIDE}([\text{Total Spend}], [\text{Total Conversions}], 0)$

Both divide the SAME numerator (Total Spend) by a different denominator, depending on which “unit” of results you’re measuring cost against — cost per click vs. cost per actual conversion. CPA will always be a larger number than CPC, since conversions are a smaller, harder-to-reach subset of clicks.

Measure 10: Return on Ad Spend (ROAS) — the headline metric

Return on Ad Spend (ROAS) = $\text{DIVIDE}([\text{Total Revenue}], [\text{Total Spend}], 0)$

Why this is the single most important measure in the project: every other funnel metric (CTR, Conversion Rate, CPC, CPA) measures EFFICIENCY at getting people through the funnel — but ROAS is the only measure that answers the actual business question: **“Was the money spent worth it?”** A channel could have a great CTR and Conversion Rate but still lose money if the cost per conversion exceeds the revenue each conversion generates — ROAS is what catches that.

Formatting note: ROAS should be displayed as a plain decimal (e.g., “4.45” or “4.45x”), NOT as a Percentage — a ROAS of 4.45 means \$4.45 returned per \$1 spent, which is a multiplier, not a proportion of a whole.

Why We Picked These Exact Charts

- One important number (Spend, ROAS, CTR) → **Card**. A single fact, kept big and simple.
- Showing people dropping off at each stage (views → clicks → sales) → **Funnel chart**. It’s the one chart type built specifically to show things shrinking step by step — a bar chart can’t show that “narrowing” feeling as clearly.
- Comparing return-on-spend across platforms → **Bar chart, biggest to smallest**. Easiest way to rank things side by side.
- Comparing spend vs revenue across platforms → **Two bars per platform, side by side**. This makes the GAP between what was spent and what came back obvious at a glance, instead of having to compare two separate charts in your head.

PHASE 5: Report Build — What Was Built, What Went Wrong, and Why

Final report structure

- Title/Subtitle, 5 KPI Cards (Spend, Revenue, ROAS, Conversion Rate, CTR)
- Funnel chart: Impressions → Clicks → Conversions
- Bar chart: ROAS by Channel (sorted descending)
- Combo chart: Spend and Revenue by Channel
- 4 Slicers: Channel, Campaign Type, Device, Region
- Insight callout + credit line

Issue 1: Funnel chart built with fields in the wrong wells

What happened: The first attempt at the funnel visual rendered as a single distorted shape with no visible stage labels, instead of three clean, narrowing bands.

Root cause: The three measures (Total Impressions, Total Clicks, Total Conversions) needed to all sit together in the **Values** field well, in that specific top-to-bottom order — not split across Category/Legend and Values, which is what happened initially.

Fix: Re-dragging all three measures into the Values well as a single ordered list corrected the chart into a proper 3-stage funnel shape.

Issue 2: Funnel showing “0M” for Clicks and Conversions despite correct underlying data

What happened: Even after fixing the field placement, the funnel displayed Total Impressions correctly (12M) but showed both Clicks and Conversions as “0M” — despite the Cards elsewhere on the page already proving these numbers were correct (460K and 24K respectively).

Root cause — a display/formatting issue, not a calculation error: The funnel visual’s **Display Units** setting was set to “Millions.” Since 460,000 and 24,000 both round down to “0” when forced into a millions-scale label, the visual was technically showing accurate numbers, just formatted in a way that made two of the three stages unreadable.

Fix: Format pane → Values → Display Units → changed from “Millions” to a scale that shows each number meaningfully (Auto or Thousands) — after which the funnel correctly displayed 11,646K → 460K → 24K.

The general lesson: when a visual shows an unexpected “0” for a value you’ve already independently verified is correct elsewhere (like on a Card), check the DISPLAY formatting (rounding, display units, decimal places) before assuming the underlying calculation is broken. This is the same category of issue as the HR project’s 0.22 vs 22% Attrition Rate card — DAX calculating correctly, but the visual’s formatting misrepresenting it.

Issue 3: Insight callout left with unfilled placeholder text

What happened: The insight callout template was pasted in with a literal bracketed placeholder — “[lowest-performing channel]” — still sitting in the text instead of being replaced with the actual channel name (LinkedIn Ads, confirmed from the ROAS by Channel chart).

The general lesson: when reusing a written template or suggested phrasing, always do a final read-through specifically checking for unfilled placeholders before considering a dashboard finished — this is an easy, common mistake when copy-pasting suggested text quickly.

Final verified numbers from the built report

Chart	Finding
ROAS by Channel	Google Ads highest, LinkedIn Ads lowest
Spend vs Revenue by Channel	Every channel shows revenue exceeding spend, but by varying margins 11,646K impressions → 460K

Why the insight callout was worded the way it was

“Google Ads delivers the highest ROAS among all channels, while LinkedIn Ads trails behind despite comparable spend — suggesting budget reallocation toward Google Ads could improve overall efficiency.”

This callout does more than state a fact — it draws an actionable business RECOMMENDATION (reallocate budget) from the finding, rather than just describing the data. Explicitly connecting a data finding to a suggested action is a stronger analytical move than a purely descriptive statement, and is exactly the kind of insight a marketing stakeholder would want highlighted first.

Key Takeaways From This Project

1. **The same repetition rule for extracting dimension tables applies across every domain** — Channels here, Plans in the SaaS project, Departments in the HR project — all follow identical reasoning: extract when a value repeats across many fact-table rows.
2. **Sequencing cleanup steps correctly prevents bugs before they happen** — removing blanks/duplicates BEFORE adding custom columns (learned the hard way in NorthStar) was applied proactively here rather than discovered as an error.
3. **Funnel-style metrics use DIFFERENT denominators at each stage** — CTR divides by Impressions, Conversion Rate divides by Clicks, not the funnel’s original top-of-funnel number every time. Each stage’s rate is relative to the stage before it.
4. **When a visual shows an unexpected “0” or blank for a number you’ve already verified elsewhere, check display formatting (units, decimal places, rounding) before assuming the calculation itself is broken** — this saved significant debugging time compared to assuming the DAX was wrong.
5. **ROAS is the metric that actually answers “was this worth it,”** while CTR/Conversion Rate/CPC/CPA only measure efficiency getting people through the funnel — a channel can look efficient at every earlier stage and still be a poor investment if ROAS is low.
6. **Always do a final read-through for unfilled template placeholders** before considering written content (like an insight callout) finished.
7. **The strongest insight callouts connect a finding to a specific recommended action**, not just a restated fact — turning “Channel X has low ROAS” into “reallocating budget toward Channel Y could improve efficiency” is a more valuable, stakeholder-ready statement.